Our Universe

A Layperson's Hypothesis

Author: Andrew Robus
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Hypotheses

- There is only one elementary particle: the energy quantum.
- Our Universe is a black hole.
- Universe = E

Particle Physics

Level	Example
Observation	Two photons produce an electron-positron pair.
Mathematical description	Energy, momentum, and conservation equations
Standard interpretation	Quantum field theory
My interpretation	Energy quanta reconfigure themselves into an electron and a positron.

The Black Hole

1. Basic Assumption

Our Universe is a black hole and is therefore embedded in another universe, which is also a black hole. Consequently, everything that exists (all universes combined) forms one single gigantic black hole. All existing energy accumulates in one place and appears as a black hole.

2. Formation and Comparison

- A star collapses into a black hole.
- The formation of the event horizon creates a one-way causal boundary around an inner region of spacetime, from which no signals can escape to the outside. This region remains part of the overarching spacetime, but its interior develops independently. For an observer inside this region, the beginning of this independent development represents the origin of their cosmological history.

For comparison, we use Sagittarius A*, with a mass of 8.54×10^{36} kg and a radius of 1.27×10^{10} m.

From these values, we calculate a simplified geometric mean density of approximately 9.98×10^{5} kg/m³. We adopt this value for the black hole that forms our Universe.

When our original object (e.g. a collapsed star) collapsed into a black hole, this was our Big Bang.

When we calculate the history of our Universe backwards toward the Big Bang, we are effectively calculating back to the singularity. However, it is possible that this goes too far back and that such a state never actually existed.

We assume that we were a black hole with an average density of approximately 9.98×10^5 kg/m³ and a diameter of 90 light-years (corresponding to approximately 21 seconds after the Big Bang according to the Standard Model).

- **Time:** 21 seconds after the Big Bang
- **Diameter:** Approximately 90 light-years
- **Average density:** Approximately 9.98×10^5 kg/m³

All of this is purely speculative and refers to our observable Universe, which is not necessarily its entire extent.

Expansion

We were therefore a densely filled sphere with a diameter of at least 90 light-years. This size has not changed.

Our own gravity continues to act on us. What standard physics describes as the curvature field, calculated from the total mass and the center of mass of a system, is, when expressed in terms of energy equivalents, an energy field that exerts pressure on energy.

This means that everything (ultimately all energy quanta) shrinks uniformly, depending on its distance from the center, making space appear increasingly larger to us.

The fact that we do not observe a preferred direction in the Universe could be due either to its size or to the possibility that, as everything shrinks, the center moves away from us at exactly the same speed (or possibly faster) than everything moves toward it.

It could therefore appear to us as though we are expanding outward, even though everything continues to become more concentrated toward the center, while the center itself shrinks faster than we can follow.

Radiation pressure from the early phase of the Universe may also have given the total mass an impulse toward the outer boundary of the black hole, which continues to counteract the inward pull of gravity.

If the compression of the strings toward the mass center of the black hole accelerates due to the increasing gravitational influence, this could also be consistent with accelerated expansion. What we observe in our Universe would then simply be increasingly diluted filaments rather than the strings themselves.

The Size Relationships of Our "Parent Universe"

If we assume that gravity compresses us and therefore causes us to shrink, we must reverse this process to look back toward our parent universe.

This universe is one level above us. It experiences much less energetic pressure than we do, which means that its elementary particles themselves are compressed much less than ours.

Because everything there is much larger, their universe appears just as large to them as ours appears to us. However, they have much more mass and space than we do.

Imagine all energy and mass gathered in one place. This forms a gigantic black hole with a gigantic universe.

Within this gigantic universe, there are black holes, each of which also contains a universe of its own. However, these consist only of a collapsed star from the perspective of the gigantic parent universe.

If we were to fly into a black hole in our Universe, we would enter a daughter universe of our own. It would contain much less matter than ours and would also have less space.

Its space would also expand, but it would have started much smaller and later than ours. It would be a miniature universe compared to ours.

Dark Matter

In this hypothesis, we assume that the observed gravitational effects currently attributed to dark matter could potentially be explained by the structure and dynamics of the black hole whose interior forms our Universe.

Two possible mechanisms are to be investigated initially.

1. Radial Compression of the String Structure

The energy of the strings moves toward the mass center of the black hole and becomes increasingly concentrated there.

This could change the gravitational influence of the mass center on the distribution of matter within the system.

In particular, it would need to be investigated whether this compression could produce a gravitational effect that persists despite increasing spatial distance from the mass center.

One possible approach would be a time-dependent distribution of the enclosed mass and energy.

2. Rotation of the String Structure and Spacetime

Since the black hole in this hypothesis does not exist outside our Universe but instead represents its underlying structure, the rotation of its strings could simultaneously mean a rotation of the underlying spacetime.

The Kerr spacetime provides a mathematical starting point for this idea.

It describes a rotating black hole and takes into account the influence of its angular momentum on spacetime geometry.

A well-known effect is frame dragging (the Lense-Thirring effect), in which the rotation of a black hole causes local inertial frames to be dragged along with it.

It would be worth investigating whether a corresponding rotating string structure could influence the motion of matter within the inner universe and potentially contribute to gravitational effects currently attributed to dark matter.

Perhaps it could also explain why our galaxies do not all move toward a common center, but instead maintain stable orbits that counteract the gravitational pull.

3. Further Investigation

The two approaches do not necessarily exclude one another. It is conceivable that radial compression and rotation work together.

Possible mathematical starting points include the Kerr spacetime, the Einstein field equations, and the investigation of radial mass and energy distributions.

In particular, it would need to be examined whether these mechanisms could reproduce the observed flat rotation curves of galaxies and other gravitational effects without assuming an additional dark matter component.

At present, these remain open hypotheses. Whether these mechanisms are physically possible and can reproduce the observations must first be investigated mathematically.

Open Question: Matter-Antimatter Asymmetry

If our Universe originated from the collapse of an already existing star that consisted predominantly of matter, it must be investigated whether and in what form the matter-antimatter problem of standard cosmology arises in this model.

In particular, it must be examined how the existing matter is preserved during its transition into the proposed energy quantum and string structure.

Could the Parent Universe Consist of Antiparticles?

Do the antiparticles fall into the black hole and fail to convert back into energy quanta, instead remaining as positrons and the other antiparticles corresponding to our particles?

Inside the black hole, there could be a kind of bounce before or during the Big Bang, causing all particles to fly backward.

This would mean that everything in our daughter universes consists of the antiparticles corresponding to the particles in our parent universe.

Notes

For simplicity, we have used the terms curvature and center of mass.

However, since this belongs to the hypothesis "Universe = E", these terms would eventually need to be replaced by concepts such as energy field and energy center.

Additional Hypotheses: String Dynamics, Hawking Radiation and Fluctuations

1. Relationship Between Internal Compression and Hawking Radiation

The strings form the fundamental structure of the black hole and extend radially from one boundary to the other (alternatively, from the boundary to the center).

I would rather suspect that the strings are arranged similarly to a chain extending from the boundary toward the center (not as individual large strings, but as many small ones).

Their energy becomes increasingly concentrated toward the mass center.

External perspective

The increasing energy concentration could be related to continuous energy radiation at the outer boundary of the black hole.

There may be a connection to Hawking radiation.

Internal perspective

The same change in string energy density could appear as the shrinking of our matter and as the formation or accumulation of gas and radiation around galaxies.

Both processes would represent different perspectives on the same underlying dynamics.

Long-term development

In the long term, the black hole could evaporate through energy radiation.

At the same time, our Universe would become increasingly dilute and cool.

As the black hole disappears, its internal universe structure could also lose the basis for its existence and eventually dissolve into the vacuum of the overarching universe.

2. Fluctuations Through Interaction with Infalling Matter – Possibility A

An object approaching the black hole is increasingly affected by spaghettification until it is broken down into its elementary particles (at the latest when crossing the event horizon).

These particles then spread across a large area of the black hole.

This could appear as a temporary particle fluctuation.

3. Fluctuations Through Interactions Between Strings – Possibility B

Another possibility is that the strings themselves oscillate and interact with one another.

- The strings could transfer energy and momentum between one another.
- Their interactions could produce local changes in energy density.

- From the internal perspective, these changes could appear as the temporary emergence or disappearance of particles or particle configurations.

4. Open Questions

- Are strings actually capable of oscillating?
- How is absorbed energy distributed within a string?
- Can interactions between strings produce temporary particle fluctuations?
- How can these processes be reconciled with known quantum-mechanical fluctuations and conservation laws?
- Is there a common mathematical relationship between expansion (or our shrinking), hot gas in galactic halos, and Hawking radiation?

Additional Mathematical Notes

5. Relationship Between Energy Quantum Size and the Overall Energy Field

A mathematical relationship is needed between the size of an energy quantum and the local overall energy field.

A first abstract representation would be:

$l_q(r,t) = f(\Phi_E(r,t))$

Where:

- **l_q:** Characteristic size of an energy quantum
- **Φ_E:** Strength of the proposed overall energy field
- **r:** Distance from the energy center
- **t:** Temporal development

This is not yet a physical equation, but a representation of what the hypothesis would mathematically require.

6. Additional Hypothesis: Minimum Size and Compression Limit

Energy quanta may possess a minimum physical size that cannot be reduced further by the overall energy field.

Once this limit is reached, further compression toward the energy center may no longer be possible.

An increasing internal pressure could then trigger a dynamic transition (bounce).

Questions to investigate

- Does a minimum size of energy quanta exist?
- What mechanism prevents further compression?
- Can the resulting counterpressure cause a reversal of the dynamics?

- How are the bounce, the Big Bang, and subsequent cosmological development connected?

7. Helicity Reversal Through a Bounce

Consider the following thought experiment:

1. You move forward.
2. You rotate around your own axis to the left.
3. You bounce off a wall.
4. Your direction of movement reverses.
5. Your rotational direction remains unchanged.

The projection of the spin onto the new direction of movement is therefore reversed.

The helicity is defined by the projection of spin onto the direction of momentum.

If the momentum vector reverses direction while the spin vector remains unchanged, the sign of the helicity changes.

This raises the question of whether an extreme bounce inside a black hole or during the Big Bang could cause a more fundamental transformation of particle configurations.